

# AL-AZHARI LIBRARY — Backend Phase 9 Implementation Report

Phase 9 complete.  
Implementation stopped here.  
No Phase 10 Shipping & Coupons work was implemented.

## 1. Executive Summary & Verification Metrics

Phase 9 (**Payments**) has been successfully implemented, verified, and integrated into the AL-AZHARI LIBRARY backend. All requirements ( `PAY-001` through `PAY-008` ), architectural constraints, direct-upload Cloudinary signing, transactional proof submission, admin review workflows, optimistic concurrency versioning, audit logging, outbox event persistence, and database migrations are fully validated.

### Exact Pipeline & Regression Baseline Results

Typecheck: PASS (tsc --noEmit - 0 errors)  
Lint: PASS (eslint . - 0 errors, 0 warnings)  
Build: PASS (tsc - dist build clean)  
Test Suites: 81 passed, 81 total (74 Phase 8 baseline + 7 Phase 9 suites)  
Tests: 563 passed, 563 total (0 failures, 0 regressions)  
Snapshots: 0 total

## 2. PAY-001 – PAY-008 Requirements Traceability

| ID      | Requirement Statement                                                                                 | Implementation Details                                                                                                                                                                                                                                             | Status    |
|---------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| PAY-001 | Support InstaPay and approved electronic wallets (Vodafone Cash, Orange Cash, Etisalat Cash, WE Pay). | Configuration-driven registry in <code>payment-methods.config.ts</code> , validating enabled methods during checkout and order creation.                                                                                                                           | CONFIRMED |
| PAY-002 | Customer upload of one or more payment screenshot proofs.                                             | <code>POST /api/v1/orders/:reference/payment-proofs</code> accepts 1 to 5 uploaded image metadata objects.                                                                                                                                                         | CONFIRMED |
| PAY-003 | No automated/external payment gateways; no customer transaction ID inputs.                            | Fully manual proof workflow. No payment gateway SDKs (Stripe, PayPal, Fawry API). No transaction reference inputs from customers.                                                                                                                                  | CONFIRMED |
| PAY-004 | Admin review actions: Confirm, Reject, Request New Proof.                                             | <code>POST /admin/payments/:id/confirm</code> , <code>POST /admin/payments/:id/reject</code> , and <code>POST /admin/payments/:id/request-new-proof</code> .                                                                                                       | CONFIRMED |
| PAY-005 | Rejection moves order to <code>awaiting_new_proof</code> , customer can re-upload new proof.          | Rejection/new proof request sets order to <code>awaiting_new_proof</code> , payment to <code>rejected</code> or <code>new_proof_requested</code> , allowing subsequent proof submission with incremented <code>submissionNumber</code> . Old proofs remain intact. | CONFIRMED |
| PAY-006 | Store uploaded payment screenshots via approved provider (Cloudinary).                                | Direct-upload with constrained signature scoped to <code>al-azhari/payment-proofs</code> . Stores only metadata in <code>paymentProofs.files[]</code> . Private assets delivered via temporary signed URLs.                                                        | CONFIRMED |
| PAY-007 | Maintain comprehensive audit trail of payment-related actions.                                        | Appended via centralized <code>AuditService</code> inside transaction ( <code>payment.proof_submitted</code> , <code>payment.confirmed</code> , <code>payment.rejected</code> , <code>payment.new_proof_requested</code> ).                                        | CONFIRMED |
| PAY-008 | No internal audit detail exposed to customers beyond transparency needs.                              | Customer projection ( <code>toSafeCustomerPaymentResponse</code> ) returns sanitized status, safe submission number, and customer notes; internal review details and actor IDs are excluded.                                                                       | CONFIRMED |

## 3. Implemented Architecture & Components

### 3.1 Domain Models & Collections

#### 1. `payments` Collection ( `payment.model.ts` )

- Schema fields: `ownerType` ( `order`, `serviceQuotation` ), `ownerId`, `customerId`, `methodKey`, `methodSnapshot`, `amountDueMinor` (integer EGP minor units), `currency`, `status` ( `not_submitted`, `under_review`, `confirmed`, `rejected`, `new_proof_requested` ), `proofRequired`, `proofSubmissionCount`, `confirmedAt`, `rejectedAt`, `metadata`, `version`.
- Indexes:
  - Unique compound index on { `ownerType`: 1, `ownerId`: 1 }
  - Status queue index on { `status`: 1, `updatedAt`: 1 }

- Customer query index on { customerId: 1, createdAt: -1 }
- 2. **paymentProofs Collection** ( payment-proof.model.ts )
  - Schema fields: paymentId , ownerType , ownerId , customerId , submissionNumber (sequential integer counter), files[] ( cloudinaryPublicId , resourceType: 'image' , format: 'png'|'jpeg'|'jpg'|'webp' , bytes , width , height , sha256 ), status , customerNote , reviewNote , reviewedBy , reviewedAt .
  - Indexes:
    - Unique compound index on { paymentId: 1, submissionNumber: 1 }
    - Status review queue index on { status: 1, createdAt: 1 }
    - Owner query index on { ownerType: 1, ownerId: 1, createdAt: -1 }
- 3. **outboxEvents Collection & Service** ( outbox-event.model.ts and outbox.service.ts )
  - Schema fields: eventType , aggregateType , aggregateId , payload , dedupeKey , status ( pending ), attempts , availableAt , processedAt , lastError .
  - Outbox events created transactionally: payment\_submitted , payment\_verified , payment\_rejected , payment\_proof\_requested .

## 3.2 State Machine Separation

- **Order State Machine:** pending\_review —► awaiting\_payment (Admin accept) —► payment\_verification (proof upload) —► payment\_confirmed (Admin confirm) or awaiting\_new\_proof (Admin reject/request new proof).
- **Payment State Machine:** not\_submitted —► under\_review —► confirmed / rejected / new\_proof\_requested .
- Confirmed is a terminal financial state ( confirmed cannot transition to any other status).

## 3.3 Cloudinary Direct Upload & Private Asset Policy

- generatePaymentProofUploadConfig : Generates a constrained signature strictly bounded to folder al-azhari/payment-proofs , resourceType: 'image' , and formats ['png', 'jpeg', 'jpg', 'webp'] . Never leaks CLOUDINARY\_API\_SECRET .
- validatePaymentProofFile : Validates public ID folder prefix, resource type, format, and byte size (1 to 10MB bounds).
- generatePrivateDownloadUrl : Admin endpoint generates short-lived (300-second expiration) signed URLs for authorized proof inspection on demand. No permanent public URLs are ever persisted or exposed.

## 3.4 Concurrency & Transactional Boundaries

- Proof submission and admin review actions execute within withTransaction() sessions.
- In-transaction failure/rollback asserts that payment status, order status, proof documents, audit log records, and outbox event records are 100% rolled back.
- Optimistic concurrency control via expectedVersion guarantees that duplicate or racing review requests fail safely with PAYMENT\_VERSION\_CONFLICT OR PAYMENT\_REVIEW\_STATE\_CONFLICT .

## 4. API Endpoints Reference

| Method | Path                                                                  | Auth / Permission             | Descript                                      |
|--------|-----------------------------------------------------------------------|-------------------------------|-----------------------------------------------|
| GET    | /api/v1/orders/:reference/payment                                     | Customer JWT or X-Guest-Token | Get payn method : instructio custome history. |
| POST   | /api/v1/orders/:reference/payment-proof/upload-config                 | Customer JWT or X-Guest-Token | Get cons Cloudina upload p                    |
| POST   | /api/v1/orders/:reference/payment-proofs                              | Customer JWT or X-Guest-Token | Submit u metadat: payment under_re payment.   |
| GET    | /api/v1/admin/payments                                                | Admin ( payments.review )     | List revie status fil paginatio               |
| GET    | /api/v1/admin/payments/:paymentId                                     | Admin ( payments.review )     | Get payn and full p submissi                  |
| POST   | /api/v1/admin/payments/:paymentId/confirm                             | Admin ( payments.review )     | Confirms moves on payment.                    |
| POST   | /api/v1/admin/payments/:paymentId/reject                              | Admin ( payments.review )     | Rejects p reason; r awaitin                   |
| POST   | /api/v1/admin/payments/:paymentId/request-new-proof                   | Admin ( payments.review )     | Requests with note to awaitin                 |
| GET    | /api/v1/admin/payments/:paymentId/proofs/:submissionNumber/signed-url | Admin ( payments.review )     | Generate private s proof ins                  |

## 5. Artifacts and Documentation Created

1. docs/PHASE-9-PAYMENTS.md & Back-end/docs/PHASE-9-PAYMENTS.md
  - Complete technical documentation covering Scope, PAY-001–008 traceability, Data Models, State Machine, Cloudinary flow, Validation, Authorization, Admin review, Order integration, Audit, Outbox, Concurrency, Indexes, Migration, Security, and Open Decisions.
2. Back-end/src/config/openapi.ts
  - OpenAPI 3.0.3 specification updated with all Phase 9 routes, parameters, request bodies, and component schemas.
3. Database Migration 20260928\_007\_payments.migration.ts
  - Idempotent migration creating payments and paymentProofs collections and required blueprint indexes.

#### 4. Comprehensive Test Suites:

- `tests/api/payment-api.test.ts` : Customer/guest payment access, upload-config generation, proof submission, COD rejection, cross-customer/cross-guest isolation, folder rejection, and secret leak assertions.
- `tests/api/payment-admin-api.test.ts` : Admin RBAC ( `payments.review` ), queue pagination, confirm, reject, request-new-proof, version conflict safety, and short-lived signed URLs.
- `tests/integration/payment-proof-lifecycle.test.ts` : Full lifecycle (submit #1 -> request-new-proof -> submit #2 -> confirm), COD rejection, rejection flow, audit trail, and outbox event creation.
- `tests/integration/payment-concurrency.test.ts` : Concurrent submissions (unique `submissionNumber` ), concurrent review race (confirm vs reject), and multi-document transaction rollback.
- `tests/unit/payment-schema.test.ts` : Zod schema validations.
- `tests/unit/payment-state-machine.test.ts` : State machine transition validation and illegal transition blocking.
- `tests/unit/payments-migration.test.ts` : Migration up/down idempotency and index verification.